

Aminoglycosides

Thomas P. Lodise
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Short View Summary

- Usual intravenous (IV) adult dose for gram-negative infections: 5–7 mg/kg every 24 hours for gentamicin and tobramycin; 15 mg/kg for plazomicin; and 20–30 mg/kg every 24 hours for amikacin
- Usual dose for synergy against gram-positive infections: 3 mg/kg/day gentamicin every 8, 12, or 24 hours
- Dose reductions required in patients with chronic kidney disease and acute kidney injury (AKI) or dysfunction
- The plasma concentration–time curve to minimum inhibitory concentration (AUC/MIC) ratio is the pharmacokinetic/pharmacodynamic parameter most closely linked to bacterial killing and efficacy
- Therapeutic drug monitoring: not always necessary for brief administration (≤ 3 days); trough concentrations should be unmeasurable; typically, two serum concentrations in postdistribution phase (≥ 1 hour after administration, separated by ≥ 1.5 half-lives)
- Cerebrospinal fluid penetration: low
- Common adverse effects: AKI, ototoxicity
- Genotyping for the mitochondrial genome variant in *MT-RNR1*, m.1555A>G can help avoid use of aminoglycosides and prevent ototoxicity
- Drug-drug interactions: increased acute kidney risk with concomitant amphotericin B, cisplatin, cyclosporine, foscarnet, furosemide, IV radiocontrast agents, methoxyflurane, polymyxins, and vancomycin
- Indications: sepsis/septic shock; hospital-acquired/ventilator-associated pneumonia; complicated intraabdominal infections; urinary tract infections; neutropenic fever; acute exacerbation of pulmonary disease (cystic fibrosis patients only); bloodstream infections/IV catheter-related infections; infective endocarditis; multidrug-resistant tuberculosis; select nontuberculous mycobacterial infections; zoonotic infections (i.e., brucellosis, chronic forms of bartonellosis, tularemia, and yersiniosis); febrile neutropenia; surgical prophylaxis for patients undergoing genitourinary, gastrointestinal, or colorectal procedures; orthopedic surgery (antibiotic-impregnated cement); ocular infections (topical preparations); sexually transmitted infections; and select protozoal infections (e.g., amebiasis and cutaneous leishmaniasis)

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Aminoglycoside antibiotics have been used since the 1940s, when systematic screening of soil actinomycetes for the elaboration of antimicrobial substances yielded streptomycin. Produced by a species of *Streptomyces* (*S. griseus*), it was the first antibiotic in the aminoglycoside family (Table 24.1) ¹ to be derived, directly or indirectly, from *Streptomyces* spp. (aminoglycosides with names ending in -mycin) or from *Micromonospora* spp. (aminoglycosides with names ending in -micin). Neomycin, kanamycin, and gentamicin are fermentation products with two or three chemical constituents. Amikacin, netilmicin, and others are semisynthetic derivatives of the natural product. Plazomicin is a semisynthetic aminoglycoside derived from sisomicin. ² All aminoglycosides share similar physical, chemical, and pharmacologic properties. Four compounds—amikacin, gentamicin, tobramycin, and plazomicin—are currently approved by the US Food and Drug Administration (FDA) for the treatment of serious infections due to gram-negative or gram-positive bacteria. ³⁻⁶

The advantages of the aminoglycoside class include rapid bactericidal activity, presence of a postantibiotic effect (PAE), synergy when combined with certain antibiotics, and well-defined pharmacokinetic (PK) profiles. ⁷⁻¹¹ As polycationic molecules, aminoglycosides bind to anionic compounds on the cell membrane, such as lipopolysaccharides (LPS), and displace outer membrane divalent cations, leading to increased permeability of the membrane and uptake of aminoglycosides (i.e., self-promoted uptake). Displacement of outer membrane divalent cations by aminoglycosides also compromises the integrity of the cell wall, and it is postulated this contributes to the observed bactericidal action of aminoglycosides. ¹²⁻¹³ After entering the cell, aminoglycosides bind to the 30S component of the prokaryotic ribosome. ⁹ Affinity for the ribosomal target contributes to the rapid bactericidal action of aminoglycosides. ⁹⁻¹⁴ Aminoglycoside activity is pH-dependent and is inhibited in acidic environments. ⁷ In vitro studies have shown that aminoglycosides have a prolonged PAE, during which bacterial growth is suppressed for a further 3 to 8 hours after drug concentrations fall below the minimum inhibitory concentration (MIC). ⁷⁻⁸⁻¹¹⁻¹⁵ Aminoglycosides are highly synergic with cell wall-active drugs (i.e., β -lactams or glycopeptides) in vitro. In preclinical pharmacokinetic/pharmacodynamic (PK/PD) infection models, concomitant use of an aminoglycoside with a β -lactam often enhances bacterial killing and prevents regrowth and resistance emergence to both the aminoglycoside and companion β -lactam. ¹⁶⁻⁴⁵ The ratio of the maximum plasma drug concentration to the MIC ($C_{\max}/\text{MIC}$) ratio has long been considered the PK/PD driver for aminoglycosides, but more recent evidence suggest the area under the plasma concentration–time curve to MIC (AUC/MIC) ratio is the PK/PD parameter most closely linked to bacterial killing and efficacy. ¹⁰⁻²⁹⁻³⁶⁻⁴⁶⁻⁴⁹

Aminoglycosides are active in vitro against both gram-positive and gram-negative pathogens, but they are primarily recommended for use in combination therapy for most infections. ⁵⁰⁻⁶³ Three legacy aminoglycosides—amikacin, gentamicin, and tobramycin—are broadly active against *Staphylococcus* spp. (primarily gentamicin) and gram-negative bacteria including *Pseudomonas* spp. (primarily amikacin and tobramycin), *Citrobacter* spp., *Enterobacter* spp., *Escherichia coli*, *Klebsiella* spp., *Proteus* spp., and *Serratia* spp. ³⁻⁵ Plazomicin, a newly approved aminoglycoside, is highly active in vitro against *Staphylococcus aureus*, including methicillin-resistant (MRSA) strains, and most Enterobacterales, including carbapenem-resistant Enterobacterales (CRE) and strains resistant to ≥ 1 aminoglycosides. ⁶⁻⁶⁴⁻⁶⁵ Some aminoglycosides have useful activity against mycobacteria, paromomycin has activity against selected colonic protozoan pathogens, and spectinomycin, a related antibiotic, has activity against *Neisseria gonorrhoeae*. ⁶²⁻⁶³⁻⁶⁶⁻⁶⁸

Although aminoglycosides are effective at bacterial cell killing and have a low potential for allergic reactions, their mechanism of action can also increase the potential for certain toxicities. ⁶⁹⁻⁷² The affinity of aminoglycosides for receptors on the proximal renal tubular epithelial cells can lead to acute kidney injury (AKI). ¹¹ In addition, ototoxicity (cochlear and vestibular) and, rarely, neuromuscular blockade are potential aminoglycoside-related side effects. ⁹⁻¹⁴⁻⁷³ Since the mid-1990s, clinicians have generally adopted weight-based, extended-interval short-course dosing strategies for

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TABLE 24.1

Family of Aminoglycosides in Clinical Use

carbapenem-resistant Enterobacterales (CRE) and strains resistant to ≥ 1 aminoglycosides. ^{6, 64, 65} Some aminoglycosides have useful activity against mycobacteria, paromomycin has activity against selected colonic protozoan pathogens, and spectinomycin, a related antibiotic, has activity against *Neisseria gonorrhoeae*. ^{62, 63, 66-68}

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TABLE 24.1

Family of Aminoglycosides in Clinical Use

GENERIC NAME	PROPRIETARY NAME	SOURCE	YEAR REPORTED	CHEMISTRY
Streptomycin	None	<i>Streptomyces griseus</i>	1944	Unique central aminocyclitol ring
Neomycin	Mycifradin, Neobiotic	<i>Streptomyces fradiae</i>	1949	Roughly equal proportions of neomycin B and C
Kanamycin	Kantrex	<i>Streptomyces kanamyceticus</i>	1957	Mixture of 95% kanamycin A and 5% kanamycin B
Paromomycin	Humatin	<i>S. fradiae</i>	1959	Part of neomycin family
Spectinomycin	Trobicin	<i>Streptomyces spectabilis</i>	1961	Chemically distinct but closely related to aminoglycosides
Gentamicin	Garamycin	<i>Micromonospora purpurea</i> and <i>Micromonospora echinospora</i>	1963	Roughly equal proportions of gentamicin C ₁ , C _{1a} , and enantiomers C ₂ and C _{2a}
Tobramycin	Nebcin	<i>Streptomyces tenebrarius</i>	1967	Natural 3'-deoxy derivative of kanamycin B
Sisomicin ^a	Siseptin	<i>Micromonospora inyoensis</i>	1970	Dehydro analogue of gentamicin C _{1a}
Dibekacin ^{a, b}		<i>S. kanamyceticus</i>	1971	Dideoxy derivative of kanamycin B
Amikacin ^b	Amikin	<i>S. kanamyceticus</i>	1972	Semisynthetic derivative of kanamycin A

Kanamycin	Kanulex	<i>Streptomyces kanamyceticus</i>	1957	Mixture of 50% kanamycin A and 5% kanamycin B
Paromomycin	Humatin	<i>S. fradiae</i>	1959	Part of neomycin family
Spectinomycin	Trobicin	<i>Streptomyces spectabilis</i>	1961	Chemically distinct but closely related to aminoglycosides
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Amikacin ^b	Amikin	<i>S. kanamyceticus</i>	1972	Semisynthetic derivative of kanamycin A
Netilmicin ^b	Netromycin	<i>M. inyoensis</i>	1975	N-Ethyl derivative of sisomicin
Isepamicin ^{a, b}		<i>M. purpurea</i>	1978	I-N-S- α -Hydroxy B amino propionyl derivative of gentamicin B
Plazomicin	Zemdri	<i>M. inyoensis</i>	2018	Semisynthetic derivative of sisomicin

^a Approved for human use in countries other than the United States.

^b Semisynthetic aminoglycosides.

Modified from Wright GD, Berghuis AM, Mobashery S. Aminoglycoside antibiotics: structures, functions, and resistance. In: Rosen BP, Mobashery S, eds. *Resolving the Antibiotic Paradox: Progress in Understanding Drug Resistance and Development of New Antibiotics*. New York: Plenum; 1998.

Amid concerns about toxicity (particularly irreversible vestibular injury), the use of aminoglycosides began to decline in the 1980s in favor of newer antibiotic classes such as fluoroquinolones, which were perceived to be less toxic. ⁹ The 2021 National Safety Network Antimicrobial Use Option Report indicates that amikacin, tobramycin, and gentamicin account for 1% to 2% and 2% to 10% of adult and pediatric antibiotic prescriptions in the inpatient setting, respectively. ⁷⁶ However, the increasing prevalence of multidrug-resistant (MDR) gram-negative pathogens, including CRE, *Pseudomonas aeruginosa*, and *Acinetobacter* spp., and nontuberculous mycobacteria—for which therapeutic options are limited—has led to renewed interest in aminoglycosides for use as monotherapy or in combination with other antibacterials. ^{7, 66, 67, 72, 77} Although most studies have failed to demonstrate improved outcomes in patients treated with antibiotic combinations with an aminoglycoside relative to monotherapy, ^{78, 79} patients with septic shock receiving combinations including